

CRANBROOK, BC, Canada

WMO: 716600

Lat: 49.6175N

Lon: 115.7892W

Elev: 930

StdP: 90.64

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-22.5	-19.3	-27.1	0.4	-21.2	-23.5	0.5	-19.0	8.0	1.5	7.2	1.7	0.8	0	0.597

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.7	31.8	16.3	30.0	15.7	28.0	15.1	17.5	28.1	16.6	27.1	15.7	25.9	4.0	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.9	11.1	19.4	12.7	10.3	19.0	11.6	9.5	18.7	52.7	28.2	49.8	27.1	47.2	26.1	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.0	7.0	6.1	-26.9	34.6	4.0	2.0	-29.8	36.1	-32.1	37.3	-34.4	38.4	-37.3	39.9		
			-27.1	19.2	4.0	1.4	-30.0	20.2	-32.3	21.0	-34.5	21.9	-37.4	22.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.3	-5.7	-3.6	1.8	6.4	11.4	14.9	19.5	18.7	13.2	5.6	-1.0	-6.4
	DBStd	10.04	6.58	5.68	4.48	3.52	3.86	3.49	3.45	3.32	3.96	4.05	5.29	6.03
	HDD10.0	2273	488	382	256	118	29	3	0	0	14	146	331	508
	HDD18.3	4531	746	615	513	358	216	114	28	37	159	396	581	767
	CDD10.0	913	0	0	1	9	73	149	295	269	109	8	0	0
	CDD18.3	129	0	0	0	0	1	11	65	48	5	0	0	0
	CDH23.3	2000	0	0	0	1	58	185	912	716	128	0	0	0
	CDH26.7	692	0	0	0	0	7	48	353	255	29	0	0	0
Wind	WSAvg	2.2	1.5	1.8	2.5	2.7	2.8	2.7	2.7	2.6	2.3	2.1	1.8	1.4
Precipitation	PrecAvg	408	36	24	20	29	45	52	33	36	34	19	35	40
	PrecMax	542	86	60	34	64	120	156	95	87	100	46	92	101
	PrecMin	291	4	4	6	3	11	8	6	0	0	3	4	10
	PrecStd	60	23	14	8	18	28	34	19	22	25	12	24	23
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.4	9.3	15.7	22.5	27.5	31.1	34.5	33.6	29.8	20.5	11.9	6.4
		MCWB	4.1	4.6	7.1	11.2	13.8	16.1	17.1	16.4	14.9	12.3	7.8	3.5
	2%	DB	5.4	6.6	12.4	19.1	25.0	27.6	32.4	31.4	27.0	16.9	9.2	4.5
		MCWB	2.8	2.8	5.7	9.6	12.8	14.9	16.6	16.0	14.3	10.1	6.3	2.4
	5%	DB	3.9	5.0	10.1	16.2	22.4	25.3	30.6	29.7	24.2	14.4	7.3	3.0
		MCWB	1.8	1.9	4.7	8.0	11.8	14.0	16.2	15.3	13.4	8.8	4.7	1.4
	10%	DB	2.4	3.5	8.2	13.6	19.7	22.8	28.5	27.6	21.6	12.2	5.7	1.5
		MCWB	0.7	0.9	3.7	6.7	10.6	13.2	15.5	14.9	12.5	7.5	3.5	0.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.7	5.2	8.1	11.8	14.8	17.5	19.1	18.1	16.4	12.9	8.8	4.4
		MCDB	6.7	7.9	13.1	20.6	25.0	28.5	30.7	28.8	25.7	19.3	11.3	5.7
	2%	WB	3.1	3.4	6.5	10.0	13.6	15.8	17.9	17.0	14.9	10.9	6.6	2.7
		MCDB	4.8	5.9	11.3	18.0	22.5	25.3	28.4	28.0	25.0	15.8	8.9	4.1
	5%	WB	2.0	2.2	5.2	8.5	12.4	14.7	17.0	16.2	13.9	9.3	5.1	1.5
		MCDB	3.7	4.5	9.3	15.1	20.7	23.5	27.5	27.0	23.0	13.7	6.8	2.9
	10%	WB	0.8	1.1	4.1	7.2	11.3	13.7	16.1	15.4	12.9	7.9	3.8	0.2
		MCDB	2.3	3.2	7.7	12.8	18.6	21.2	26.5	25.7	20.9	11.6	5.4	1.4
Mean Daily Temperature Range	5% DB	MDBR	7.3	9.3	9.9	11.6	12.4	12.3	14.7	15.2	13.5	10.6	7.2	6.5
		MCDBR	7.7	9.5	12.8	16.6	16.9	16.3	17.8	18.4	18.2	13.8	8.6	7.0
		MCWBR	5.7	6.5	7.5	8.8	8.0	7.1	6.7	7.1	8.2	7.8	6.1	5.4
	5% WB	MCDBR	7.2	8.6	11.6	14.9	15.0	14.5	15.9	16.4	16.8	12.2	8.0	6.6
		MCWBR	5.5	6.1	7.1	8.1	7.4	6.7	6.4	6.8	7.8	7.2	6.0	5.3
Clear-Sky Solar Irradiance	taub	0.256	0.263	0.286	0.351	0.361	0.361	0.358	0.370	0.336	0.298	0.273	0.249	
	taud	2.358	2.349	2.370	2.264	2.354	2.403	2.423	2.397	2.472	2.565	2.514	2.393	
	Ebn at Noon	811	896	931	893	896	897	895	866	865	851	795	771	
	Edh at Noon	64	83	98	123	117	113	109	107	89	68	56	54	
All-Sky Solar Radiation	RadAvg	1.05	2.00	3.00	4.42	5.28	5.64	6.65	5.51	3.83	2.26	1.10	0.82	
	RadStd	0.11	0.18	0.32	0.37	0.45	0.48	0.54	0.42	0.41	0.27	0.12	0.08	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (33 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page